

Geography Resources and Development

Important - Terms

1. Resource - Everything which can be used to satisfy our needs
2. Three things needed for resources -
(i) Technologically accessible
(ii) Economically feasible
(iii) Culturally acceptable
3. Total geographical area of India -
3.28 million sq km / km²
4. The degraded land present in India -
130 million hectares
5. Two factors on which development depends
Technology
Quality of human resource
6. Two states where over-irrigation is responsible for land degradation.
- Punjab & Haryana

7. According to age alluvial soil divided into - Khadar and Bhangar
8. Two minerals that make alluvial soil more fertile.
- Potash and lime
9. A crop of alluvial soil - wheat
10. Cotton soil is known as - Regur or black soil
11. A soil which has a good capacity to hold moisture - black soil
12. A crop of black soil - cotton
13. A soil which develops ^{by} crystalline igneous rocks - red soil
14. Red soil looks yellow - when it's hydrated
15. A soil which develops with high temperature & heavy rainfall - laterite soil
16. Two crops of laterite soil - Tea and coffee

17. The most widely spread soil in India
Alluvial soil (40%)
18. Two states which are well endowed with solar energy - Rajasthan and Gujarat
19. Bad land in Chambal basin called
ravines or gully
20. A resource which is found but has not utilized - Potential resource, Planting lines of trees to create shelter belts upto the force of wind is known as shelter belts
21. A state with water resource but lacks in infrastructure development - Arunachal Pradesh
22. States rich in minerals and coal deposits -> Jharkhand, Chhattisgarh, Madhya Pradesh
23. Two global ecological crisis - global warming, ozone layer depletion

25. The cold desert of Ladakh - lacks in water resources and some vital minerals
26. A land not for cultivation - waste or barren land
27. Removal of top soil is known as - soil erosion

Short Notes :-

1. Agenda 21
 - It was adopted at first international Earth Summit held in 1992 at Rio de Janeiro
 - It has focused to achieve global sustainable development
 - Its aim is to look environmental damage, poverty and disease through global co-operation
 - Under it, every local government should draw its own Agenda 21
2. Resource Planning
 - It is a complex process as it involves identification and keep record of resources with survey, mapping and

qualitative and quantitative estimation and measurement.

- For the planning structure there is proper technology skill and institutions are set up.
- To compare resources, development plans should come with overall national development plans.

3) # Enormous diversity of Resources but deficient in some other resources :-

We know in India there are different types of resources but we agree with given statement because :-

1. Arunachal Pradesh has abundance of water resources but lacks in infra-structural development.
2. While Rajasthan is well endowed with solar and wind energy but lacks in water resources.
3. Other side Ladakh (cold desert) has rich cultural resource but difficult to get minerals resources.

Land

It is a natural resource of utmost important as we live on land and perform our economics activities. It supports systems of life.

Relief features of land

- I. Plains (43%) - All settlements, crop farming and other diverse human activities are found in plains.
- II. Mountains (30%) - In it we found, forest, wildlife, river system, etc.
- III. Plateaus (27%) - It contains minerals, forest, and some seasonal rivers.

Land utilisation

- I. Forest
- II. Land not for cultivation
 - (i) Barren & wasteland
 - (ii) Land put to use for buildings, roads, factories, etc.
- III. Uncultivated land

- (ii) Pastures land
- (iii) Miscellaneous trees
- (iii) Cultivable land (for 5 year)

4. Fallow lands
- (i) Current fallow lands
 - (ii) Current fallow lands (for 1-5 years)
 - (iii) Net sown area - total gross cropped area

Conservation of Land

Degradation of resource

Some human activities have contributed significantly in land degradation are as follows -

- (i) Deforestation - due to mining, farming and other settlements have caused severe land degradation.
- (ii) Over-irrigation & use of chemicals - have become a source of it.
- (iii) Industrial waste & Mineral process - like grinding of lime stone, generates harmful dust particles that settles down on the land

and degrades it.

Steps To control land degradation:-

- (i) Afforestation is the best way to conserve it.
- (ii) Proper farming techniques like crop rotation, terrace farming should adapt.
- (iii) Restricted grazing areas should be moved from different pasture areas to avoid soil erosion.

Soil

- (i) It is a basic resource and used for agriculture.
- (ii) In other ways soil is disintegrated rock materials mixed with humus and forming the outermost layers of the crust.

Features:-

- It is loose & fragmented
- It consists of both organic and inorganic substances

* Main constituents of soil

sand, silt and clay govern the texture of soil.

Types of soil:

1) Alluvial soil:

- These soils are formed by river system as they bring silt & minerals in their long journey.

The main contents of the soil is lime and potash.

- They are deficient by organic and nitrogen contents.

It is very fertile, covers 30% area of India.

- These are found in northern plains and coastal strips.

- The main crops of the soil are rice, wheat, oil seeds, sugarcane, etc.

2) Black soil:

- It is the result of weathering of volcanic (basalt) rock formed by the deccan lava.
- It is rich in lime, potash, magnesium, Iron, calcium.
- It lacks by nitrogen and humus.
- It is self ploughing quality, holds moisture when wet.
- These are found in north western parts of the deccan plateau. Crops grown are - cotton, sugarcane.

3) Red and Yellow soil

- These are formed due to decomposition of metamorphic rocks (igneous rocks).
- The main content of the soil is iron.
- This soil is deficient by nitrogen, phosphorus and lime.
- They do not retain moisture so needs fertilizers.
- They are found in parts of madhya Pradesh, odisha, TN.
- They look yellow when hydrated.
- wheat, rice, cotton, pulses can grow in it.

4. Laterite Soil

- These soils are formed due to leaching.
- It is rich in iron but lacks in phosphorus, lime, nitrogen, etc.
- It is acidic, porous, and porous by nature.
- This type of soil is found mostly in western and eastern ghats and some part of north eastern states like Assam and Meghalaya.
- Coffee, rubber, cashews and the main crops of the soil.

5. Desert Soil / Sandy soil

- It is formed due to mechanical weathering of rocks from sands.
- It is very loose, porous and deficient by organic matter.
- It has high proportion of salts.
- These are found in the states of Rajasthan, Gujarat and Punjab.
- With irrigation methods - wheat, Bajra, Moring, grams can grow.

6. Mountain Soil

- These are found by the weathering of rocks.
- These soils have rich organic and mineral substances.
- It is sandy, gravel and heterogeneous in nature but lacks in humus.
- These are found in hills regions of Arunachal Pradesh, J&K, HP and UK.
- These are suitable for fruits and vegetables.

#1 Soil Erosion & Conservation:-

- Removal of the top soil by (or) top layer of the soil, wind, water, is erosion.
- The processes of soil formation and erosion go on simultaneously and generally there is a balance between the two.

Soil Conservation Methods:-

- Sheet erosion
- Terrace farming
- Shelter farming
- Contour ploughing

#2 Removal of top layer of the soil is called soil erosion. For this some agent, wind, water and air are working and responsible for its movement.

Types of soil erosion:-

- I. Sheet erosion → It takes place when
 - II. Gully erosion → running water
 - III. Wind erosion → the top layer
- It occurs when water cuts deep channels
- It happens when winds blow away loose soil

① Contour Ploughing - This method is using to slow down the water flow, that helps to reduce erosion.

② Terrace Farming - In this, steps are cut, on slopes that prevent soil from being washed away.

③ Strip Farming - under it land is divided into strips of crops and grass which are the

wind force.

- 4) Shelter belts - reduce wind erosion, rows of trees are planted because it breaks wind speed
- 5) Planting more trees to bind soil is the best method to prevent soil erosion

National Forest Policy (1952):-

1995-26-

A common future a book under sustainable development introduced — by Brundland Commission (1987)

Small is beautiful (1974) → Schumacher (Gandhian philosophy)

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24/12